Uncertainty-Aware Contrastive Learning for Semi-Supervised Classification of Multimodal Remote Sensing Images

Kexin Ding, Ting Lu, *Member, IEEE*, and Shutao Li, *Fellow, IEEE*

***Abstract*— Recently, deep learning (DL) presents a promising performance in the joint classification of multimodal remote sensing (RS) data. However, most of the approaches adopt a supervised learning (SL) manner, where the discrimination capability is limited by the paucity of labeled samples. Though some attempts have been made to develop semi-supervised methods, they prefer to select the highly confident predictions as pseudo-ground truth and discard those unreliable ones. Actually, unreliable samples can also provide useful information, e.g., indicating the categories to which samples may belong and definitely not belong. Focused on this, a novel uncertainty-aware contrastive learning (UACL) method is proposed. Here, label uncertainty analysis (UA) based on multilevel probability estimation is first conducted to separate reliable and unreliable samples, which are then processed with a designed hybrid ("hard" or "soft") contrastive learning (CL) strategy. For reliable samples, the "hard" CL pushes the network to learn features that will minimize the intra-class distance while maximizing the inter-class distance, according to the pseudo-labels. For unreliable samples, the "soft" CL aims to learn the similarity and difference among samples, where the predicted class probabilities are queried to estimate a soft mask for an adaptive feature similarity measurement. Moreover, a multimodal spectral–spatial joint feature representation pipeline of triple branches, i.e., one spectral branch for hyperspectral images (HSIs) and two spatial branches for multimodal data, is also introduced. By jointly learning from both labeled and unlabeled samples, more discriminative spectral–spatial feature representation will lead to a further boost in classification performance. Extensive experiments on four well-known multimodal datasets prove the effectiveness of the proposed semi-supervised classification method. Codes are available at https://github.com/Ding-Kexin/UACL.**

***Index Terms*— Contrastive learning (CL), hyperspectral images (HSIs), light detection and ranging (LiDAR) data, multimodal remote sensing (RS) data, semi-supervised classification, synthetic aperture radar (SAR).**

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I. Introduction

CLASSIFICATION of remote sensing (RS) images is essential for many applications, including urban planning, forest management, environmental monitoring, and precision agriculture [1], [2], [3], [4]. With the continuous development of RS imaging technologies, a large amount of multimodal data is available [5], [6], [7], [8]. Among them, hyperspectral images (HSIs) are frequently applied to accurate land cover classification because they contain both rich spatial and spectral information simultaneously [9], [10], [11]. Light detection and ranging (LiDAR) and synthetic aperture radar (SAR) data record the elevation information of ground objects, which is not only resistant to interference but also provides complementary information to hyperspectral data [12], [13]. Consequently, classification approaches that focus on the fusion of HSIs and LiDAR/SAR data have been widely investigated and significantly advanced [8], [14].

Over the past few years, deep learning (DL) models have gained a lot of interest in multimodal image fusion and classification tasks thanks to their efficient capabilities of feature learning [15], [16], [17]. Among these methods, the majority adopts a supervised learning (SL) manner, i.e., aiming at discovering general and discriminative features from limited labeled samples. For example, Xu et al. [18] first used a two-branch convolutional neural network (CNN) to extract the spectral–spatial features of the HSIs, and then stacked them with the spatial CNN features from LiDAR to perform fusion classification. In [19], a hierarchical random walk layer was proposed to achieve spatially consistent optimization of classification boundaries. In [20], multiscale attention mechanisms were introduced to learn spectral–spatially enhanced representations of features from HSIs and LiDAR data. In [21], a multimodal vision transformer (ViT) was introduced to learn multisource integration features through information fusion of patch tokens. Nevertheless, despite the efforts of existing DL-based joint HSI and LiDAR/SAR classification schemes in feature extraction and optimization of classification results, their performance is inevitably affected by the limited number of labeled samples.

To overcome the dependence on labeled samples, a series of unsupervised learning (UL) approaches based on DL have been presented in the RS image classification area. In [22], the network learns the relationship from the HSI to the LiDAR

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directly at the image level, expecting the features used for classification to have HSI and LiDAR data fusion information. In [23], an unsupervised feature extraction framework was developed to accomplish the mapping from one modality to another. By integrating multiscale features from two different sources in the hidden layer, deeply fused features are obtained that can be used for classification. Xu et al. [24] introduced a multibranch capsule network to combine features from HSI and LiDAR with semantical comprehension to achieve unsupervised extraction and recognition of spectral–spatial–elevation hybrid features. The above methods attempt to solve the labeled sample dependency problem, but the lack of ground truth supervision leads to less than optimal classification performance.

To effectively exploit both the labeled and unlabeled samples to enhance classification results, some semi-SL (SSL) strategies are developed, which have been emerging as a research hotspot. These methods aim to quench the thirst for labeled sample size while guaranteeing better classification results. One typical method is the generative model-based sample augmentation, which aims at increasing dataset size via data generation/augmentation. In [25], a generative model was introduced to generate more hyperspectral samples similar to the real data, in this way, the small sample problem is effectively alleviated. In [26], a conditional generative adversarial network (GAN) based on clustering was presented to increase the size and quality of the training dataset. Another well-known SSL method is the graph-based label propagation, which attempts to construct the correlation between labeled and unlabeled samples to guide label propagation and label prediction processes. In [27], the construction of an adjacency matrix from HSIs and LiDAR data emphasizes the correlation between multimodal pixels, thus improving semi-supervised classification performance. In [28], an updatable graph is constructed from high-level features, allowing cross-modal label propagation. In overall, the performance of the generative model-based sample augmentation methods is highly dependent on the quality of the samples generated and sometimes not stable. The graph-based label propagation methodologies often involve huge computational overheads, which somewhat limits the performance of the models.

In addition, another widely developed SSL method is the pseudo-label-based self-training, which tries to exploit the prediction results of unlabeled samples along with the learning process. These methods are popular since numerous advanced supervised DL models can be transferred to construct the semi-supervised model via introducing an effective training strategy, meanwhile, the performance improvement compared to supervised methods is always obvious [29], [30], [31], [32]. For example, in [29], a simple but effective form of pseudo-labeling approach was offered. The way it works is that the maximum prediction probability category is chosen as the pseudo-label, which is used as the true label, thus achieving low-density separation between classes through entropy regularization. Subsequently, SSL models based on fixed or dynamic thresholds [30], [31] have been used widely depending on how to assess the confidence level of the predicted pseudo-labels. Wu and Prasad [32] treated the clustering labels of unlabeled samples as pseudo-labels and applied them to the pre-training of deep convolutional recurrent neural networks, thus improving the classification model initialization and enhancing the classification performance. Moreover, traditional self-training approaches mostly utilize only reliable or high-confidence pseudo-labels, neglecting the useful discriminative information of hard samples.

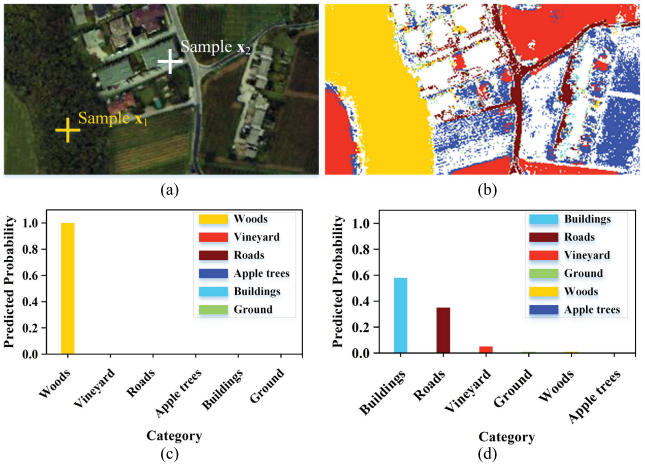


Fig. 1. Illustration of the pseudo-label reliability. (a) False-color image for HSI of the Trento dataset (using 33, 16, and 4 for R, G, and B, respectively). (b) Colored pseudo-labels correspond to reliable predictions of easy samples, while pixels within the white region have no reliable pseudo-labels assigned to them. (c) Predicted probability distribution corresponding to a reliable pseudo-label (i.e., sample $\mathbf{x}_1$), which should rightly be regarded as a positive sample of *Woods*. (d) Predicted probability distribution corresponding to an unreliable pseudo-label (i.e., sample $\mathbf{x}_2$), which can be considered positive samples of *Buildings* and *Roads* to varying degrees. More importantly, there is a general belief that they are not part of any other category and can be considered definitive negative samples.

On the basis of the above analysis, we attempt to design a new uncertainty-aware contrastive learning (UACL) methodology for multimodal RS data semi-supervised classification. The major inspiration is to extensively explore the properties of a large number of unlabeled samples in the feature learning process, thereby alleviating the reliance on labeled sample quantities. Different from the commonly used highly reliable sample selection strategies in existing pseudo-label-based SSL methods, both the reliable and unreliable unlabeled samples remain and are used adaptively here. This is reasonable since the unreliable samples can also provide useful discriminative and similar information, as explained in Fig. 1. In specific, the proposed UACL is divided into two parts: multimodal spectral–spatial joint representation (MSJR) and pseudo-label uncertainty-aware contrastive learning (PUCL). The first part is dedicated to jointly training for more discriminative spectral–spatial feature extraction and high-precision classification, with the effective exploit of both labeled and unlabeled sample information from multimodal data. In the second part, unlabeled samples are separated into easy and hard samples by considering the deep prediction probability distribution and the consistency of the shallow and middle feature classification results. Subsequently, a hybrid contrastive learning (HCL) strategy is designed for processing easy and hard samples differently. The UACL model is based on a self-training scheme where only the training set is subject to cross-entropy loss in the early training phase, while

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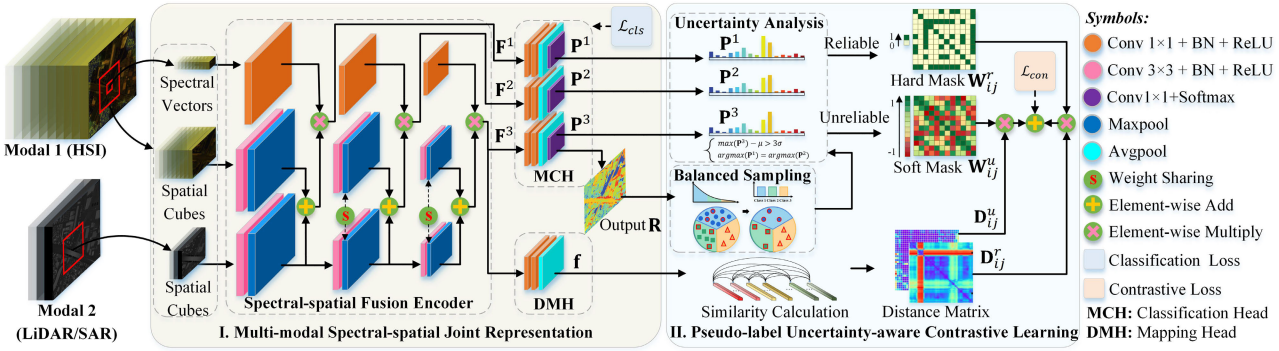


Fig. 2. Proposed framework of UACL for semi-supervised classification of multimodal RS images.

pseudo-label contrastive loss is added in the later training period to achieve robust semi-supervised classification. The proposed method is shown to outperform other state-of-the-art methods both objectively and subjectively through extensive experiments on four well-known HSI and LiDAR/SAR datasets. The main contributions of this work are outlined below.

1) A new method for semi-supervised multimodal RS image fusion classification driven by HCL is proposed. The method unifies the multimodal feature joint learning task and the pseudo-label CL task in one framework. In this manner, both tasks will cooperate to push the network to learn more discriminative feature representation from both labeled and unlabeled samples, leading to the classification performance improvement.
2) To take full advantage of the multimodal complementary information, a new MSJR pipeline with three branches is proposed. Here, the spectral–spatial features from different data are adaptively fused while the multilevel (shallow, middle, and deep) features are also jointly used. In this manner, more discriminative features can be effectively learned for class categorization.
3) To reduce the dependence on the labeled sample volume, a PUCL strategy for the fusion classification of HSI and LiDAR/SAR data is proposed for the first time. It constructs an uncertainty analysis (UA) criterion and utilizes all unsupervised information for differentiated learning, i.e., "hard" CL for reliable samples and "soft" CL for unreliable samples, which enhances the efficiency of unlabeled sample utilization.

The remaining work is arranged as follows. Section II presents the specifics of the suggested framework. Section III provides the experimental datasets, performance, and study of different methods. At last, the conclusion is discussed in Section IV.

II. Proposed Method

As shown in Fig. 2, the proposed UACL is composed of two parts: MSJR and PUCL. The MSJR aims at learning discriminative spectral–spatial feature representation based on both labeled and unlabeled samples. Here, a triple-branch DL structure is designed, where spectral–spatial information in multimodal data is integrated via an adaptive feature fusion and the shallow/middle/deep features are jointly exploited in a decision fusion manner. For the PUCL part, all the selected samples with pseudo-labels are divided into "reliable" and "unreliable" sets by the designed UA first. Then, a hybrid ("hard" or "soft") CL is introduced to process both sets. Overall, by optimizing an overall loss function consisting of supervised classification loss and unsupervised contrastive loss, UACL can learn more discriminative features from a small number of labeled samples and a large number of unlabeled samples. Details of the MSJR, PUCL, and the proposed overall loss function are illustrated below.

A. Multimodal Spectral–Spatial Joint Representation

To effectively fuse both the similar semantic information and complementary feature information in multimodal heterogeneous data, we built an MSJR, as shown in Part I in Fig. 2. There are three modules in the MSJR: a spectral–spatial fusion encoder (SFE), a multilevel classification head (MCH), and a deep mapping head (DMH). Briefly, SFE first linearly weights the multimodal spatial features at shallow/middle/deep levels, followed by multilevel multiplication operations on the fused spatial features and spectral features. MCH enables differentiated classification probability estimation based on limited training samples and numerous unlabeled samples. DMH aims at mapping the fused spectral–spatial feature into deeper feature space, where the similarity and difference among pseudo-labeled samples can be better preserved. They are described in more detail as follows.

1) Spectral-SFE: The SFE employs a triple-branch structure, i.e., two spatial branches and one spectral branch. Here, the HSI cube $\mathbf{X}_h \in \mathbb{R}^{w\times w\times d_h}$, the LiDAR/SAR cube $\mathbf{X}_l \in \mathbb{R}^{w\times w\times d_l}$, and the spectral vector $\mathbf{V}_h \in \mathbb{R}^{1\times 1\times d_h}$ act as inputs, respectively, where $w \times w$ is the patch size, and d_h, d_l are the initial number of spectral bands for HSI and LiDAR/SAR.

Each spatial branch mainly contains three blocks, each of which consists of the same two layers. The first layer in turn consists of a 2-D convolution with a kernel size 3×3, a batch normalization (BN), and a ReLU activation function. In order to achieve diverse feature extraction through spectral dimensional transformations, the number of convolutional kernels in the three blocks is set to 32, 64, and 128. To reduce the computational effort and to avoid the problem of overfitting, we choose ReLU as the activation function, which is also employed in [33], [34], [35], and [36]. More importantly, we chose the above parameters to be as fair as possible against other classification methods. It is worth noting that the padding operation is employed to maintain

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a stable spatial size during the convolution calculation. The second layer in each block is the max pooling operation with a pooling window size of 2×2 and the space size is halved one by one. The latter two blocks share weights to decrease the computational overhead while allowing for better fusion of multimodal heterogeneous information with similar semantics. The spectral branch contains three layers, each with the same configuration. That is, a 1-D convolution with a kernel size 1×1, a BN, and a ReLU function. The convolution kernel count is matched with the spatial branch.

Based on the features learned in the two spatial branches, spatial feature fusion will be conducted. Assuming that the features output through each block of the spatial branch are $\mathbf{F}_h^k$ and $\mathbf{F}_l^k$, respectively, with the same size of $(w/2^k) \times (w/2^k) \times (32/2^{1-k})$, where $k \in \{1, 2, 3\}$. The following equation is used to calculate the fused spatial feature $\mathbf{F}_{\text{spa}}^k \in \mathbb{R}^{(w/2^k)\times(w/2^k)\times(32/2^{1-k})}$ at the kth level:

$$\mathbf{F}_{\text{spa}}^k = \alpha \mathbf{F}_h^k + (1-\alpha)\mathbf{F}_l^k \tag{1}$$

where α is an adjustable weighting factor that can be updated in an adaptive way by the learning process of the network. Owing to the large gap between the imaging strategies of HSI and LiDAR/SAR data, $\mathbf{F}_h^k$ and $\mathbf{F}_l^k$ are considered equally important for spatial feature extraction, so the start value of α is given as 0.5.

For spectral–spatial feature fusion, assuming that the output of the features through each block of the spectral branch is $\mathbf{F}_{\text{spe}}^k$, with the size of $1 \times 1 \times (32/2^{1-k})$. The fused spectral–spatial feature $\mathbf{F}^k \in \mathbb{R}^{(w/2^k)\times(w/2^k)\times(32/2^{1-k})}$ at the kth level is given by the following formula:

$$\mathbf{F}^k = \mathbf{F}_{\text{spa}}^k \otimes \mathbf{F}_{\text{spe}}^k \tag{2}$$

where $\otimes$ is the operator of multiplying two tensors in the manner of a broadcast.

2) Multilevel Classification Head: For the MCH, the learned multilevel features $\mathbf{F}^k$ are acted as input. The features encoded at different levels contain complementary information and contribute differently to the classification results. We design the MCH to synthesize multilevel complementary information for the classification of both labeled and unlabeled samples.

The MCH contains three structurally identical blocks for the simultaneous processing of spectral–spatial fusion features $\mathbf{F}^k$ at three levels, where each block contains four layers. The first two layers have the same composition and are formed by stacking 1-D convolution, BN, and ReLU in sequence. The quantity of convolutional kernels in the first layer in every block is defined as 16, 32, and 64, respectively, while the quantity of convolutional kernels in the second layer is defined as 16, 16, and 32, respectively. Next is an average pooling layer which globally averages the multilevel input features along the spatial dimension and outputs condensed features with a patch size of 1×1. Finally through a 1-D convolution and a softmax mapping in the fourth layer, the classification probability $\mathbf{P}^k \in \mathbb{R}^{N\times C}$ from 0 to 1 corresponding to the fused feature $\mathbf{F}^k$ at each level is obtained, which are subsequently used for multilevel cross-entropy loss. N represents the sample number in each batch and C is the count of categories. Notably, since the deeper encoding features undergo more sufficient learning, the final pseudo-label prediction and classification result $\mathbf{R}_i$ for the ith sample is taken from the classification probabilities of the deepest feature (i.e., $\mathbf{P}^3$). And $\mathbf{R}_i$ can be expressed as

$$\mathbf{R}_i = \arg\max_{c\in[1,C]} \mathbf{P}_i^3(c) \tag{3}$$

where $i \in [1, N]$.

3) Deep Mapping Head: Here, the DMH acts only on the deep spectral– spatial fusion features (i.e., $\mathbf{F}^3$) of some unlabeled samples. The DMH aims to learn the similarities and differences among unlabeled samples by mapping the features of unlabeled samples to a deeper feature space.

The DMH is made up of three layers. The first two layers have identical contents. They are built by stacking 1-D convolution, BN, and ReLU functions. The quantity of convolution kernels in the first two layers is set to 64 and 32. Similar to MCH, the third layer adopts a global average pooling operation for feature generalization to obtain a condensed feature mapping $\mathbf{f} \in \mathbb{R}^{N\times 32}$.

Through DMH, on one hand, the mapping result makes more accurate descriptions of the deeper spectral–spatial feature containing rich detail information. On the other hand, since this module is very close to the generalization of classification probabilities, it also can reflect the high-order semantic information of the multimodal data. Given that the deep mapping features $\mathbf{f}$ have these properties, it is suitable for subsequent PUCL.

B. Pseudo-Label Uncertainty-Aware Contrastive Learning

Considering the general problem of the paucity of labeled samples in RS image classification, we proposed a pseudo-label uncertainty-aware contrast learning (PUCL) strategy to fully explore unsupervised information. As shown in Part II of Fig. 2, the PUCL includes four modules: balanced sampling (BS), UA, reliable pseudo-label contrastive learning (RPCL), and unreliable pseudo-label contrastive learning (UPCL). In short, in the BS stage, an identical number of unlabeled samples are randomly selected for each category according to the pseudo-label prediction results. Then, the balanced unlabeled samples are added to the training set without considering their pseudo-labels. In the UA phase, the multilevel prediction probability distribution and the feature learning laws are considered simultaneously. The unlabeled samples are divided into highly reliable and unreliable sample sets based on a comprehensive analysis of the reliability of the pseudo-labels. Highly reliable samples are considered positive samples of pseudo-labels in RPCL. In UPCL, a soft mask is applied to the feature similarity matrix of unreliable samples to achieve more effective unsupervised information mining. The reasons for designing this hybrid CL are explained below. The "reliable" pseudo-label is likely to be correct, thus, the similarity weighting of the hard mask is designed to achieve intra-class distance minimization and inter-class distance maximization. On the contrary, those unreliable sets contain more uncertainty, the soft mask is proposed to adaptively measure

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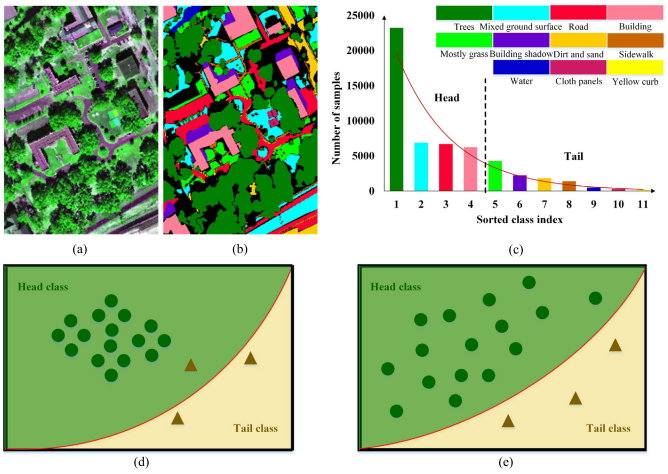


Fig. 3. Illustration of the long-tailed class distribution. (a) False-color image for HSI of the MUUFL dataset (using 20, 60, and 30 for R, G, and B, respectively). (b) Ground-truth map of the MUUFL dataset. (c) Class distribution of ground-truth samples in the MUUFL dataset, with a few head classes occupying the vast majority of samples and tail ones containing only a minority of samples. (d) Feature space before category rebalancing, where the decision boundary (i.e., red solid arcs) learned on imbalanced samples is biased toward the head class (i.e., green circle), may lead to misclassification. (e) Feature space after category rebalancing, the probability of misclassification of tail class samples (i.e., golden triangles) is reduced.

the similarities and differences among samples. The specific descriptions of the above four modules are as follows.

1) Balanced Sampling: In practice, naturally collected data classes usually have a long-tailed distribution [37], as shown in Fig. 3. That is, a few classes have a high number of samples, while most classes have a low number of samples. From the graph, it can be analyzed that the long-tailed problem may cause poor classification output, which will negatively affect SSL when a large quantity of unlabeled sample information is used. To mitigate the effects of the long-tailed distribution and improve the generalization of the UACL, a BS strategy has been adopted. Specifically, we perform pseudo-label prediction on a huge number of unlabeled samples and consider the pseudo-labels as true labels during the BS phase. Subsequently, a fixed number of unlabeled samples are randomly chosen in a category-balanced manner. Considering to minimize the computational overhead while introducing information from a large number of unlabelled samples, the number of samples selected for each category is set to be five times the number of training samples.

The evenly selected unlabeled samples are added to the training for subsequent PUCL. Through BS stage filtering, UACL alleviates the negative impact of long-tailed distribution.

2) Uncertainty Analysis: All the estimated pseudo-labels have different degrees of uncertainty, which should be treated separately. Thus, a UA criterion is proposed to separate the pseudo-labeled samples into reliable and unreliable sets first.

The inputs of UA are the multilevel classification probabilities $\mathbf{P}^k \in \mathbb{R}^{N\times C}$. Convincingly, if the maximum category probability can be regarded as an outlier, i.e., much higher than the other category probabilities, then the pseudo-label corresponding to it is reliable. Assuming that the deep prediction probability $\mathbf{P}^3$ obeys an approximately normal distribution, the probability value of a reliable pseudo-label should lie outside the triple standard deviation interval, inspired by the PauTa criterion [38]. The pseudo-labels that meet the above probability distribution criterion can be given by the following formula:

$$\hat{y}_i^{\text{pro}} = \begin{cases} \underset{c\in[1,C]}{\arg\max}\ \mathbf{P}_i^3(c), & \text{if } \mathcal{M}\left(\mathbf{P}_i^3\right) - \mu > 3\sigma \\ \text{ignore}, & \text{otherwise} \end{cases} \tag{4}$$

where $i \in [1, N]$, $\mathcal{M}(\cdot)$ represents the maximum operation, μ and σ are the mean and standard deviation of $\mathbf{P}_i^3$.

In addition, we have taken more thoughtful consideration from the perspective of feature learning. If the classification results of low-level features are consistent, it is believed that the network has effectively captured the characteristics of the sample. The pseudo-labels that meet the feature learning criterion can be given by the following function:

$$\hat{y}_i^{\text{fea}} = \begin{cases} \underset{c\in[1,C]}{\arg\max}\ \mathbf{P}_i^3(c), & \text{if } \hat{y}_i^1 = \hat{y}_i^2 \\ \text{ignore}, & \text{otherwise} \end{cases} \tag{5}$$

where $\hat{y}_i^1$ and $\hat{y}_i^2$ are the pseudo-labels corresponding to the maximum probability values predicted by the low-level features $\mathbf{P}_i^1$ and $\mathbf{P}_i^2$ of the ith sample.

In summary, reliable pseudo-labels for easy unlabeled samples should meet at least one of the above UA criteria, expressed by the following equation:

$$\hat{y}_i^{\text{rel}} = \hat{y}_i^{\text{pro}} \cup \hat{y}_i^{\text{fea}} \tag{6}$$

where $\cup$ represents a union set. The rest of the pseudo-labels from unlabeled samples are considered unreliable. UA can effectively discriminate between easy and hard samples.

3) Reliable Pseudo-Label Contrastive Learning: In the unlabeled sample set, samples with reliable pseudo-labels are easy to predict and their pseudo-labels can be used as real labels. In this way, highly reliable samples can be considered positive samples for the corresponding category of pseudo-labels and negative samples for the other categories. An RPCL loss of the "hard" CL in this work $\mathcal{L}_{\text{con}}^r$ is devised to adequately extract easy sample features, as shown below

$$\mathcal{L}_{\text{con}}^r = \frac{1}{{N_r}^2} \sum_{i=1}^{N_r} \sum_{j=1}^{N_r} \left[\mathbf{W}_{ij}^r \mathbf{D}_{ij}^{r2} + \left(1 - \mathbf{W}_{ij}^r\right) \mathcal{M}\left(0, \zeta - \mathbf{D}_{ij}^r\right)^2 \right] \tag{7}$$

where N_r is the number of samples with reliable pseudo-labels and ζ is a hyperparameter that is empirically fixed at 2. $\mathbf{W}_{ij}^r$ corresponds to the hard mask in Part II of Fig. 2, and $\mathbf{D}_{ij}^r$ is the feature similarity matrix, measured by Euclidean distance. Their specific calculation method is as follows:

$$\mathbf{W}_{ij}^r = \begin{cases} 1, & \text{if } \hat{y}_i = \hat{y}_j \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

$$\mathbf{D}_{ij}^r = \left\| \mathbf{f}_i^r - \mathbf{f}_j^r \right\|_2 \tag{9}$$

where $\mathbf{f}_i^r$ is $\mathbf{f}$ of the ith sample with reliable pseudo-label. With the guidance of $\mathcal{L}_{\text{con}}^r$, the intra-class distance of samples having the same pseudo-labels is minimized, while the inter-class distance of samples having different pseudo-labels is maximized.

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4) Unreliable Pseudo-Label Contrastive Learning: Since pseudo-labels of some samples are not credible, we cannot treat them as absolutely correct identifications. Therefore, starting from the prediction probabilities themselves, the UPCL scheme, which is also called the "soft" CL in this work, is designed.

First, the deep prediction probabilities $\mathbf{P}_i^3$ of the ith sample are sorted in descending order according to the probability values to obtain the sorted probabilities $\tilde{\mathbf{P}}_i^3 = \{\mathbf{p}_{i,1}^3, \ldots, \mathbf{p}_{i,s}^3, \ldots, \mathbf{p}_{i,C}^3\}$, where $s \in [1, C]$ is the sequence number. And the corresponding prediction category label is $\hat{y}_i = \{\hat{y}_{i,1}, \ldots, \hat{y}_{i,s}, \ldots, \hat{y}_{i,C}\}$. Subsequently, the positive sample threshold δ is set based on the probability distribution, which means that the currently hard samples can be considered positive samples of the first δ categories to varying degrees. The calculation method for δ of the ith sample is as follows:

$$\delta_i = s, \quad \text{s.t.} \quad \mathbf{p}_{i,s}^3(c) < \mu \leq \mathbf{p}_{i,s-1}^3(c). \tag{10}$$

Next, we built a weight dictionary $\mathbf{d}_i \in \mathbb{R}^C$ from -1 to 1 in order to query the correlation degree of hard sample features $\mathbf{f}_i^u$, as shown below

$$\mathbf{d}_i^s = \begin{cases} \dfrac{\delta_i - s + 1}{\delta_i}, & \text{if } 1 \leq s \leq \delta_i \\ -\dfrac{s - \delta_i}{C - \delta_i}, & \text{if } \delta_i < s \leq C. \end{cases} \tag{11}$$

Then, as shown in Part II of Fig. 2, we constructed a soft mask $\mathbf{W}_{ij}^u$ for feature similarity weighting of hard samples, as shown below

$$\mathbf{W}_{ij}^u = \begin{cases} \mathbf{d}_i^s, & \text{if } \hat{y}_{i,1} = \hat{y}_{j,s} \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where $\hat{y}_{i,1}$ is the first place of the ith hard sample predicted category label, and $\hat{y}_{j,s}$ is the sth place of the jth hard sample predicted category label.

Finally, a UPCL loss is calculated as follows:

$$\mathcal{L}_{\text{con}}^u = \frac{1}{N_u^2} \sum_{i=1}^{N_u} \sum_{j=1}^{N_u} \left(\mathbf{W}_{ij}^u \mathbf{D}_{ij}^u\right)^2 \tag{13}$$

where N_u is the quantity of samples with unreliable pseudo-labels. Similar to (9), $\mathbf{D}_{ij}^u$ is also expressed by Euclidean distance, and is calculated as follows:

$$\mathbf{D}_{ij}^u = \left\|\mathbf{f}_i^u - \mathbf{f}_j^u\right\|_2 \tag{14}$$

where $\mathbf{f}_i^u$ is $\mathbf{f}$ of the ith sample with unreliable pseudo-label. To further illustrate the hard/soft masks, examples of how to estimate the weights based on (8) and (12) are shown in Fig. 4.

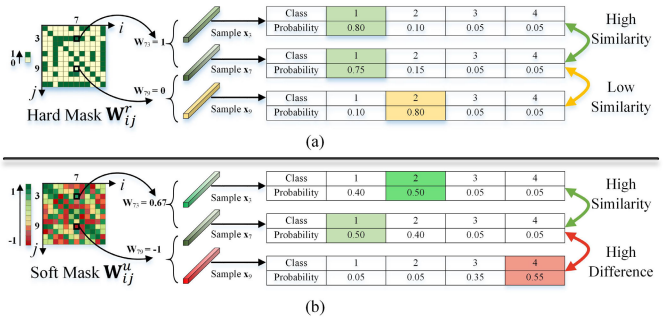


Fig. 4. Illustration of the weight calculation in hard/soft masks. (a) Hard mask estimation based on reliable pseudo-labeled samples. (b) Soft mask estimation based on unreliable pseudo-labeled samples.

C. Overall Loss Function

In the designed UACL approach, the training procedure can be divided into two parts: supervised pre-training and semi-supervised retraining. In supervised pre-training, a multilevel cross-entropy classification loss is implemented to ensure accurate classification. In the semi-supervised retraining phase, a joint loss consisting of classification loss and contrastive loss is used for constraint. The feature learning and discrimination capabilities of the model are further enhanced by the combined effect of the classification and CL tasks. These loss functions are the subject of a detailed description below.

1) Classification Loss: In Fig. 2, the classification loss $\mathcal{L}_{\text{cls}}$ acts on the multilevel classification probabilities $\mathbf{P}^k$ of the labeled samples, both during supervised pre-training and semi-supervised retraining. $\mathcal{L}_{\text{cls}}$ is calculated according to the cross-entropy loss function and can be characterized as follows:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{k=1}^{3} \sum_{i=1}^{N} \sum_{c=1}^{C} y_{i,c} \log\left(\mathbf{P}_{i,c}^k\right) \tag{15}$$

where N is the quantity of samples in one batch. For $y_{i,c}$, when the real label of the ith input cubes and vector is equal to c, $y_{i,c} = 1$, otherwise $y_{i,c} = 0$. $\mathbf{P}_{i,c}^k$ is the probability that the ith sample is predicted to be c on the kth level. $\mathcal{L}_{\text{cls}}$ helps to improve the classification accuracy.

2) Contrastive Loss: As shown in Fig. 2, the contrastive loss $\mathcal{L}_{\text{con}}$ constrains the entire PUCL process, which is a combination of reliable pseudo-label contrastive loss $\mathcal{L}_{\text{con}}^r$ and unreliable pseudo-label contrastive loss $\mathcal{L}_{\text{con}}^u$, as shown below

$$\mathcal{L}_{\text{con}} = \mathcal{L}_{\text{con}}^r + \mathcal{L}_{\text{con}}^u \tag{16}$$

where $\mathcal{L}_{\text{con}}^r$ and $\mathcal{L}_{\text{con}}^u$ are derived from (7) and (13), respectively. $\mathcal{L}_{\text{con}}$ implements unsupervised information mining for a large number of unlabeled samples, especially for hard samples that are significant for classification. Pseudo-label CL enhances the discrimination ability of the network by measuring the feature similarity of unlabeled samples, thereby improving classification performance.

3) Overall Loss: The final total loss function $\mathcal{L}$ is a linear weighting of classification loss $\mathcal{L}_{\text{cls}}$ in (15) and contrastive loss $\mathcal{L}_{\text{con}}$ in (16), expressed as follows:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \mathcal{L}_{\text{con}} \tag{17}$$

where λ is the weight factor used to equalize the classification task and the PUCL task, and its value is discussed in Section III.

In conclusion, the overall loss helps multiple tasks achieve efficient semi-supervised classification within an integrated

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framework. The entire network training procedure for UACL is described in Algorithm 1.

Algorithm 1 Training UACL Model for Semi-Supervised Classification of Multimodal RS Data

Input: Hyperspectral image $\mathbf{H}$, LiDAR or SAR data $\mathbf{L}$, training label $\mathbf{Y}$, training epochs *epochs*

Output: Pre-training classification result $\mathbf{R}_{pre}$ and final classification result $\mathbf{R}_{final}$;

Divide $\mathbf{H}$ into spatial cubes $\mathbf{X}_h$ and spectral vectors $\mathbf{V}_h$, and divide $\mathbf{L}$ into spatial cubes $\mathbf{X}_l$;
Initialize all weights;
while epoch $<$ *epochs* **do**
    Extract the multi-level encoded features $\mathbf{F}^k$ according to Eq. (2), $k \in \{1, 2, 3\}$;
    Obtain the multi-level classification probabilities $\mathbf{P}^k$ by classifying $\mathbf{F}^k$;
    Update the network according to Eq. (15) with $\mathbf{Y}$;
    Obtain the pre-training classification result $\mathbf{R}_{pre}$ by computing Eq. (3);
end while
Select unlabeled samples in a balanced manner according to $\mathbf{R}_{pre}$;
while epoch $<$ *epochs* **do**
    Analyze the reliability of pseudo-labels by solving Eq. (4), (5) and (6);
    Calculate the unlabeled sample feature similarity and mask matrix according to Eq. (8), (9), (12) and (14);
    Update the network according to Eq. (17);
    Obtain the final classification result $\mathbf{R}_{final}$ by computing Eq. (3).
end while

III. Experiments and Analysis

Here, experiments are carried out on three common HSI-LiDAR datasets and one HSI-SAR dataset to assess the performance of the designed UACL approach. First, the illustration of datasets and a brief description of the overall experimental setting will be given. Second, the influence of important parameters and the effect of important modules will be analyzed according to related experiments of parameter discussion and ablation studies. Third, to verify the superiority in terms of objective metrics and visual classification results, we also evaluate the UACL method against other advanced DL models. Here, the classification performance when varying volumes of samples are used for training is also shown, meanwhile, the computational complexity analysis is also conducted.

A. Dataset Illustration and Experimental Setting

1) Dataset Illustration: Here, the tested multimodal datasets consist of "Houston2013," "Trento," "MUUFL," and "Augsburg," and their details are shown in Table I. The corresponding land covers and number of training/test samples of each data are given in Tables II–V, respectively. The details of these four datasets can also refer to [39], [40], [41], and [42].

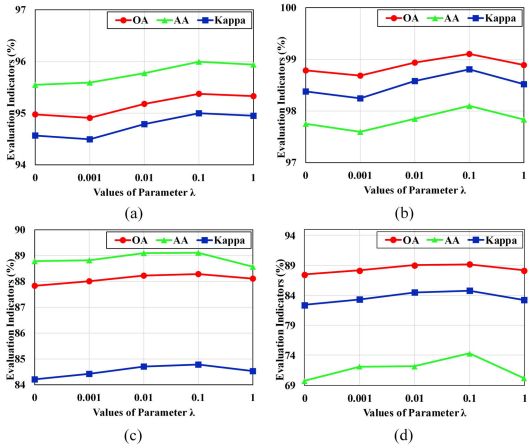


Fig. 5. Effects of the weight coefficient λ on classification performance. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

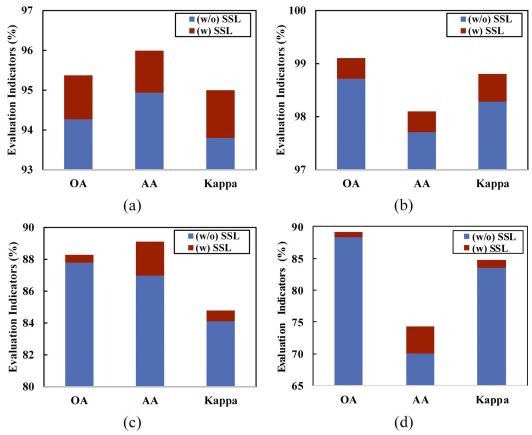


Fig. 6. Effects of the SSL on classification performance. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

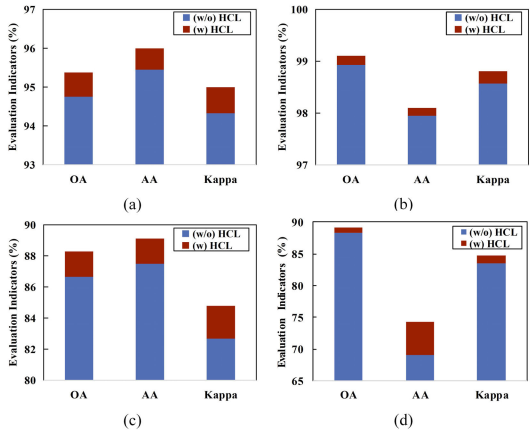


Fig. 7. Effects of the HCL on classification performance. (a) Houston2013. (b) Trento. (c) MUUFL. (d) Augsburg.

2) Experimental Setting: Our experiments run the proposed UACL approach on a server with an NVIDIA GeForce RTX 3090 GPU using the PyTorch framework. We employ an Adam optimizer with an original learning rate of $5e^{-4}$, motivated by the use of small optimization steps to achieve accurate feature learning. We have set the patch size and the batch size to 16 and 64, respectively, based on our experience.

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TABLE I
MULTIMODAL RS DATASETS USED FOR EVALUATION

Dataset Name	Houston2013		Trento		MUUFL		Augsburg	
Location	Houston, Texas, USA		Trento, Italy		Long Beach, Mississippi, USA		Augsburg, Germany	
Data Type	HSI	LiDAR	HSI	LiDAR	HSI	LiDAR	HSI	SAR
Sensor Name	CASI-1500	-	AISA Eagle	Optech ALTM 3100EA	CASI-1500	Gemini ALTM LiDAR	HySpex	Sentinel-1
Spatial Resolution	2.5 m	2.5 m	1 m	1 m	0.54 m × 1.0 m	0.60 m × 0.78 m	30 m GSD	30 m GSD
Number of bands	144	1	63	1	64	2	180	4
Wavelength	0.38 - 1.05 μm	-	0.42 - 0.99 μm	-	0.38 - 1.05 μm	1.06 μm	0.4 - 2.5μm	-
Image Size	349×1905	349×1905	166×600	166×600	325 × 220	325 × 220	332×485	332×485
Number of categories	15		6		11		7	
Number of labeled samples	15029		30214		53687		78294	

TABLE II
CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

No.	Class(Train/Test)	Performance									
		EndNet	CoupledCNN	FGCN	MAHiDFNet	TBCNN	MDL-RS	MFT	DMSGer	DSCA-Net	UACL
1	Healthy Grass(20/1231)	91.36	94.24	88.29	82.01	90.01	94.24	**95.94**	88.72	92.98	93.68
2	Stressed Grass(20/1234)	86.47	97.21	90.67	**97.94**	92.78	93.92	95.25	88.65	91.65	97.10
3	Synthetis Grass(20/677)	96.75	99.65	96.66	99.94	**99.94**	98.85	99.61	99.84	98.14	99.84
4	Trees(20/1224)	88.69	98.53	91.56	96.00	97.76	97.87	**99.32**	83.92	93.11	96.82
5	Soil(20/1222)	91.19	99.56	95.83	96.19	97.52	96.11	**99.97**	99.20	98.06	99.55
6	Water(20/305)	93.15	98.23	91.67	96.82	91.41	91.93	**100**	99.05	94.75	96.82
7	Residential(20/1248)	59.07	94.88	86.13	91.83	88.13	88.54	**96.63**	86.50	93.33	93.29
8	Commercial(20/1224)	57.79	87.38	78.23	84.35	83.53	79.84	**91.58**	61.77	81.40	87.31
9	Road(20/1232)	66.94	85.43	77.70	84.10	80.80	85.71	90.37	81.33	92.49	**93.66**
10	Highway(20/1207)	69.15	94.69	81.98	78.61	79.99	87.46	94.09	95.67	**99.50**	95.26
11	Railway(20/1215)	64.05	95.80	79.29	87.58	84.57	84.52	85.29	89.61	93.79	**96.39**
12	Park Lot 1(20/1213)	43.31	85.56	72.98	69.78	75.89	76.17	88.87	84.85	92.46	**93.52**
13	Park Lot 2(20/449)	43.92	98.60	82.47	96.17	7.88	94.28	97.62	96.37	**99.04**	97.13
14	Tennis Court(20/408)	97.55	**100**	96.03	**100**	99.41	98.09	**100**	**100**	**100**	**100**
15	Running Track(20/640)	98.61	99.69	90.16	99.97	98.50	99.09	**100**	99.70	98.41	99.56
	OA (%)	74.44	94.35	85.53	88.90	86.22	89.92	94.71	88.21	93.77	**95.37**
	AA (%)	76.08	95.30	86.64	90.75	84.54	91.11	95.64	90.34	94.61	**95.99**
	Kappa (%)	72.87	93.89	84.38	88.01	85.07	89.10	94.28	87.26	93.27	**95.00**

TABLE III
CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON THE TRENTO DATASET

No.	Class(Train/Test)	Performance									
		EndNet	CoupledCNN	FGCN	MAHiDFNet	TBCNN	MDL-RS	MFT	DMSGer	DSCA-Net	UACL
1	Apple trees(20/4014)	89.01	99.65	94.76	98.90	98.74	95.36	98.51	97.35	98.55	**99.84**
2	Buildings(20/2883)	92.04	96.60	93.48	91.15	93.42	96.76	**99.64**	87.36	96.63	98.48
3	Ground(20/459)	86.41	95.75	94.01	97.10	95.90	97.86	84.49	95.27	**98.54**	96.49
4	Woods(20/9103)	91.09	99.59	95.71	99.97	96.58	85.34	99.99	99.42	**100**	99.98
5	Vineyard(20/10481)	81.94	98.96	89.15	89.87	93.80	94.79	99.89	96.42	**100**	99.93
6	Roads(20/3154)	86.48	95.72	**92.57**	81.15	91.71	89.61	86.74	79.55	92.35	93.89
	OA (%)	87.16	98.62	92.73	93.45	95.08	91.67	98.15	94.80	98.66	**99.11**
	AA (%)	87.83	97.71	93.28	93.03	94.70	93.29	94.88	92.56	97.68	**98.10**
	Kappa (%)	83.21	98.17	90.35	91.28	93.80	89.06	97.52	93.10	98.21	**98.81**

The epochs for both supervised and semi-supervised training networks are set to 300.

For an objective reflection of the classification performance, several evaluation indicators are considered, namely overall accuracy (OA), average accuracy (AA), and Kappa coefficient. Meanwhile, the accuracy of each category is also calculated. To give a more reasonable analysis, all indicators are averaged across ten classification outcomes. In addition, CL is usually accompanied by data augmentation operations to avoid model overfitting and to improve performance, but common data augmentation steps, e.g., flipping and rotating, may be easily grasped by the model. For a fair comparison, our proposed UACL network does not employ any data augmentation operations.

B. Parameter Discussion

Here, we analyze the effect of parameter λ in (17) on classification performance, which is used to balance the contribution of the supervised classification task and the PUCL task. We fix the coefficient of the classification loss term to be constant 1, and we set the value of λ to 0, 0.001, 0.01, 0.1, and 1, respectively. A higher λ value means that the PUCL task has a higher weight, and conversely, the effect is smaller. As can be seen in Fig. 5, the influence of λ follows an analogous trend on four different datasets, i.e., too large or too small a value of λ causes a decrease in three evaluated metrics. Using the Houston2013 dataset as an illustration, when λ is set to a small value (e.g., $\lambda = 0$), the PUCL task is removed and a large amount of unlabeled sample information is discarded, limiting the classification performance. Conversely, if λ is given a large value (e.g., $\lambda = 1$), the proportion of PUCL is too high. As a result, on one hand, the importance of the supervised classification task is weakened, and on the other hand, the negative effects of incorrect pseudo-labeling may be amplified. In conclusion, an appropriate λ value is necessary for good unsupervised information exploitation and high classification accuracy. Based on the above experiments and analysis, the weighting factor λ is determined to be 0.1.

C. Ablation Studies

Here, the influence of the developed SSL framework and HCL is discussed via the ablation experiments.

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TABLE IV
CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON THE MUUFL DATASET

No.	Class(Train/Test)	Performance									
		EndNet	CoupledCNN	FGCN	MAHiDFNet	TBCNN	MDL-RS	MFT	DMSGer	DSCA-Net	UACL
1	Trees(20/23226)	79.70	89.12	83.44	79.06	84.52	85.39	89.48	41.13	88.82	**91.29**
2	Mostly grass(20/4250)	74.33	73.93	67.57	67.76	79.64	67.83	63.73	50.16	65.94	**82.06**
3	Mixed ground surface(20/6862)	61.31	58.47	53.25	64.55	65.23	54.20	61.55	39.65	69.61	**77.79**
4	Dirt and sand(20/1806)	74.11	86.36	78.56	83.65	80.66	77.30	82.67	74.40	89.09	**90.13**
5	Road(20/6667)	83.83	80.46	63.80	87.81	82.04	71.41	84.78	42.49	68.83	**88.16**
6	Water(20/446)	96.57	99.53	90.34	45.13	97.76	98.45	99.78	**99.96**	98.45	98.50
7	Building shadow(20/2213)	86.16	82.42	82.65	**92.36**	81.62	86.59	80.60	76.44	84.62	91.45
8	Building(20/6220)	90.33	**95.39**	88.74	82.01	94.23	89.55	92.36	84.63	73.16	92.91
9	Sidewalk(20/1365)	60.35	63.32	51.25	74.97	49.60	61.21	43.44	37.22	44.52	**75.71**
10	Yellow curb(20/163)	91.17	82.15	68.16	81.10	70.31	72.15	42.33	64.72	52.15	**96.07**
11	Cloth panels(20/249)	95.38	95.14	95.22	**98.07**	95.58	95.58	95.98	94.02	96.14	96.18
	OA (%)	78.50	82.69	75.52	78.15	81.45	78.01	81.80	50.18	78.93	**88.29**
	AA (%)	81.20	82.39	74.82	77.86	80.11	78.15	76.06	64.07	75.58	**89.11**
	Kappa (%)	72.78	77.70	68.97	72.45	76.27	71.90	76.37	42.42	72.69	**84.78**

TABLE V
CLASSIFICATION ACCURACIES (%) OF DIFFERENT METHODS ON THE AUGSBURG DATASET

No.	Class(Train/Test)	Performance									
		EndNet	CoupledCNN	FGCN	MAHiDFNet	TBCNN	MDL-RS	MFT	DMSGer	DSCA-Net	UACL
1	Forest(20/13487)	77.68	96.81	85.97	**98.84**	95.26	96.90	91.95	88.86	94.04	93.66
2	Residential Area(20/30309)	55.72	92.38	68.30	86.75	56.67	81.08	**93.00**	62.00	91.55	90.70
3	Industrial Area(20/3831)	35.43	42.31	59.90	34.24	44.33	50.61	51.60	66.75	**76.01**	68.91
4	Low Plants(20/26837)	31.39	91.40	32.22	31.98	85.12	61.02	88.43	57.93	76.74	**92.38**
5	Allotment(20/555)	70.54	83.24	74.77	66.56	**90.54**	90.54	58.02	90.31	74.95	56.58
6	Commercial Area(20/1625)	41.53	37.97	43.57	37.29	40.05	45.08	56.43	**62.89**	40.86	47.63
7	Water(20/1510)	65.99	63.97	43.11	52.25	42.26	57.28	61.85	**73.43**	62.32	70.26
	OA (%)	50.17	88.61	57.61	65.61	72.11	74.29	87.61	65.91	84.39	**89.19**
	AA (%)	54.04	72.58	58.26	58.27	64.89	68.93	71.61	71.74	73.78	**74.30**
	Kappa (%)	38.87	83.93	45.91	53.37	64.01	66.03	82.45	56.86	78.27	**84.80**

1) Effect of SSL: One of the starting points of UACL is to alleviate the dependence of the model on the labeled sample volume. To verify the effectiveness of SSL in helping to refine classification accuracy, we design an ablation experiment. The corresponding experimental results for three indicators (i.e., OA, AA, Kappa) on four datasets are described in Fig. 6. The blue bar and red bar indicate the result without SSL and the boost with SSL, respectively. In the absence of SSL, we only use labeled training samples to predict the classification results. The addition of SSL significantly improves the three metrics across the four datasets, as can be seen in Fig. 6. In particular, the Houston2013 dataset shows almost identical scale improvements on all three indicators. This is because its spatial size is large and the proportion of labeled samples is low. The introduction of discriminative and correlated information in numerous unlabeled samples is crucial for improving in classification performance and is reflected in all metrics. It can be summarized that SSL has indeed made a positive contribution to the joint task of classifying HSI and LiDAR/SAR.

2) Effect of the HCL: Another consideration of UACL is to measure the similarity and difference relationships of sample features through HCL, thus making the model more discriminative. To validate the positive impact of HCL on improving recognition accuracy, an ablation is designed. The corresponding experimental results for the three indicators (i.e., OA, AA, Kappa) of four datasets are depicted in Fig. 7. The blue and red bars indicate the results without HCL and the improved performance with the addition of HCL. If HCL is not added, in order to maintain an SSL strategy, we only use cross-entropy loss constraints for labeled and unlabeled samples. After adding an HCL scheme, pseudo-labels are processed in greater detail, with the reliability of pseudo-labels being considered and the deep mapping of unlabeled samples being effectively mined simultaneously. It is worth noting that for a fairer comparison, the experiments before and after ablation are based on the unlabeled samples after BS so as to alleviate the impacts of the long-tailed distribution in Fig. 3. As shown in Fig. 7, all three metrics have increased when using HCL in all four datasets. There has been a significant improvement in each indicator, particularly for the MUUFL dataset. On one hand, the long-tailed distribution in this dataset is relatively obvious, and the UA of the rebalanced unlabeled sample pseudo-labels is more necessary. On the other hand, the designed PUCL strategy allows for more diverse exploitation of deep mapping features, further improving classification performance. On the basis of the above discussion of the results from the ablation experiments, HCL can achieve differential mining of unlabeled samples and can contribute positively to classification tasks.

D. Performance Comparison

Sufficient comparison experiments are carried out for the purpose of verifying the effectiveness of the proposed UACL model. The comparative approaches contain four specialized models designed for specific modalities (e.g., HSI and LiDAR), i.e., EndNet [33], CoupledCNN [35], FGCN [36], and MAHiDFNet [43], and three multimodal generalized models, i.e., TBCNN [18], MDL-RS [34], and MFT [21]. In addition, two semi-supervised classification models, i.e., DMSGer [44] and DSCA-Net [45], are used for comparison, and it is worth noting that the multimodal images are stacked

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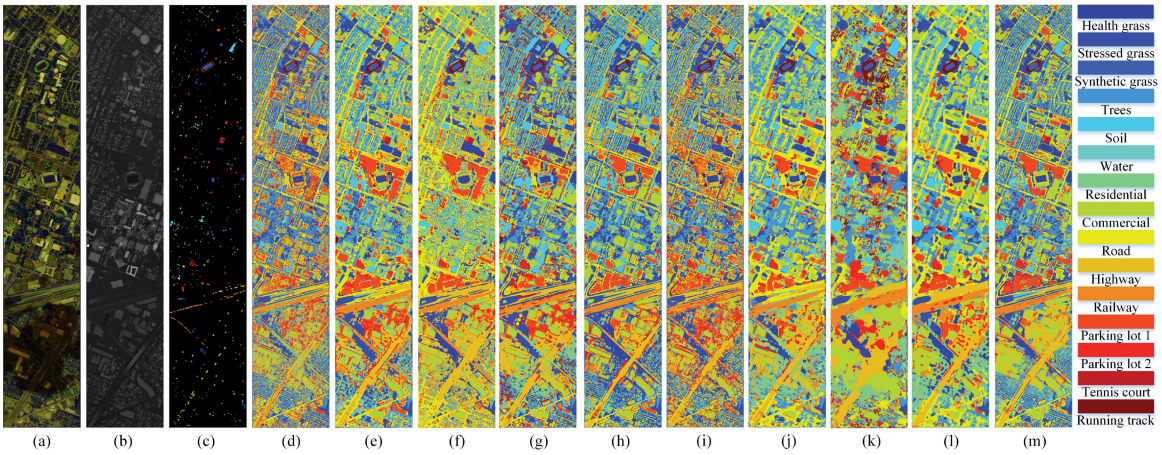


Fig. 8. Classification maps for the Houston2013 dataset obtained with different methods. (a) False-color image for HSI (using 30, 70, and 120 as R, G, and B, respectively). (b) LiDAR. (c) Ground-truth map. (d) EndNet (74.44%). (e) CoupledCNN (94.35%). (f) FGCN (85.53%). (g) MAHiDFNet (88.90%). (h) TBCNN (86.22%). (i) MDL-RS (89.92%). (j) MFT (94.71%). (k) DMSGer (88.21%). (l) DSCA-Net (93.77%). (m) UACL (95.37%).

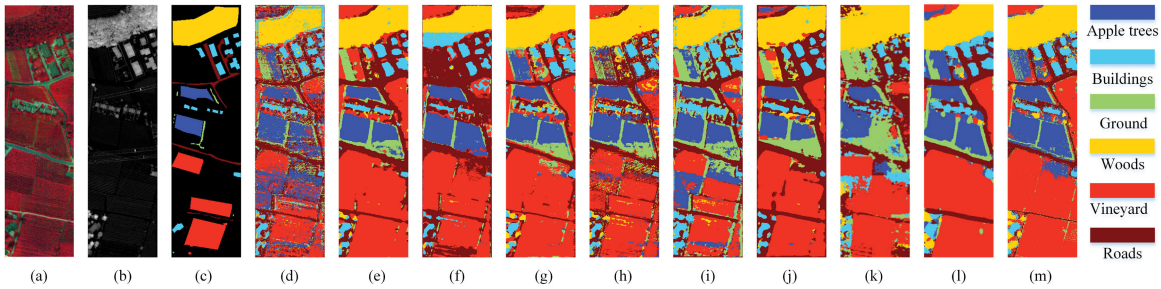


Fig. 9. Classification maps for the Trento dataset obtained with different methods. (a) False-color image for HSI (using 40, 10, and 20 as R, G, and B, respectively). (b) LiDAR. (c) Ground-truth map. (d) EndNet (87.16%). (e) CoupledCNN (98.62%). (f) FGCN (92.73%). (g) MAHiDFNet (93.45%). (h) TBCNN (95.08%). (i) MDL-RS (91.67%). (j) MFT (98.15%). (k) DMSGer (94.80%). (l) DSCA-Net (98.66%). (m) UACL (99.11%).

along the spectral dimension. For fairness, we use identical samples for training and testing on the same server. Besides, the parameter optimization of each of the techniques compared has been performed as described in the source publications or with the default settings of the open source code.

1) Comparison of Different Classification Methods: From the quantitative analysis and subjective visualization perspective, this article discusses the UACL method and compares it with the nine advanced DL-based methods for the joint classification of multimodal RS data. In Tables II–V, the sample allocation and objective indicator scores for all models on four datasets are listed, respectively. In all these tables, the highest values in each row are marked in bold. Figs. 8–11 show in turn the false-color images of the HSIs, visualizations of the LiDAR/SAR data, ground-truth diagrams, and different method classification maps on the four experimental datasets. We also display the colors used for each category.

Table II gives the OA, AA, Kappa, and classification accuracy for each category on the Houston2013 dataset. The suggested UACL approach achieves the highest value for a total of three objective evaluation indicators and obtains the greatest recognition precision for four of the 15 classes of ground objects. In comparison with the second-best method (MFT), the OA, AA, and Kappa of UACL increase by 0.66%, 0.35%, and 0.72%, respectively. For this scene, some areas (such as the lower part of the HSI false-color image in Fig. 8) are obscured by clouds and fog, which may lead to incorrect classification of ground objects (such as railways). From the visual results in Fig. 8, it can be observed that our UACL

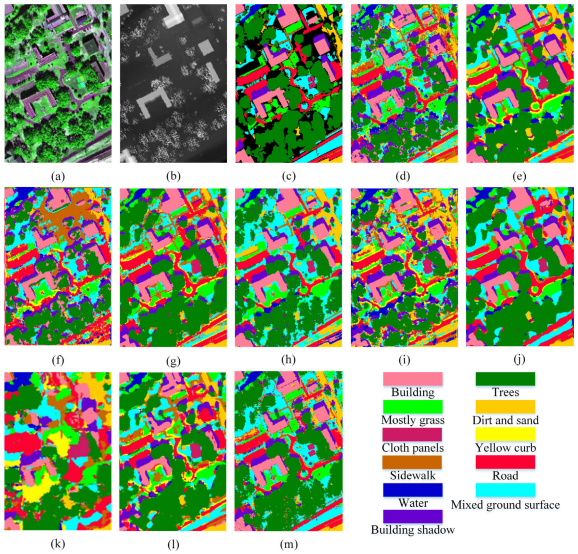


Fig. 10. Classification maps for the MUUFL dataset obtained with different methods. (a) False-color image for HSI (using 20, 60, and 30 as R, G, and B, respectively). (b) LiDAR. (c) Ground-truth map. (d) EndNet (78.50%). (e) CoupledCNN (82.69%). (f) FGCN (75.52%). (g) MAHiDFNet (78.15%). (h) TBCNN (81.45%). (i) MDL-RS (78.01%). (j) MFT (81.80%). (k) DMSGer (50.18%). (l) DSCA-Net (78.93%). (m) UACL (88.29%).

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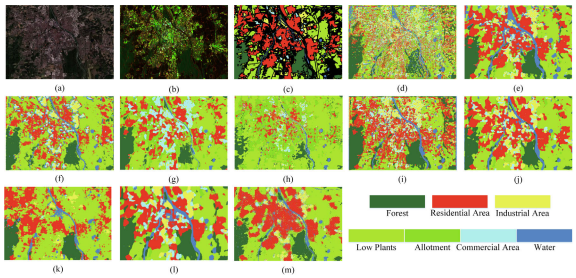


Fig. 11. Classification maps for the Augsburg dataset obtained with different methods. (a) False-color image for HSI (using 20, 15, and 10 as R, G, and B, respectively). (b) SAR. (c) Ground-truth map. (d) EndNet (50.17%). (e) CoupledCNN (88.61%). (f) FGCN (57.61%). (g) MAHiDFNet (65.61%). (h) TBCNN (72.11%). (i) MDL-RS (74.29%). (j) MFT (87.61%). (k) DMSGer (65.91%). (l) DSCA-Net (84.39%). (m) UACL (89.19%).

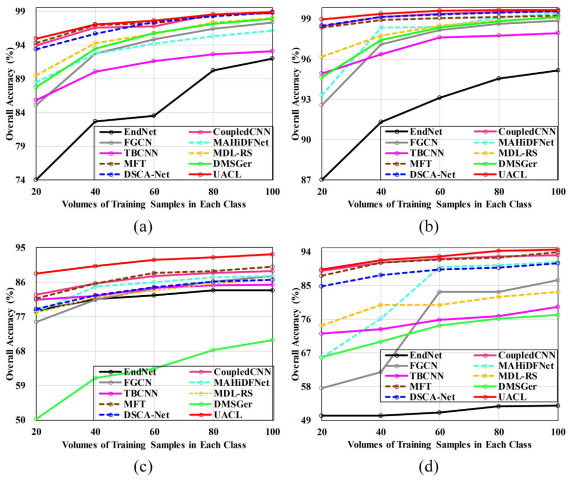


Fig. 12. Classification performance with different volumes of training samples using (a) Houston2013; (b) Trento; (c) MUUFL; and (d) Augsburg.

approach has clear boundaries and good continuity for the classification results of railways, which is inconsistent with the performance in Table II. This suggests that UACL can exploit elevation information from LiDAR/SAR more effectively to mitigate the above problems. Specifically, MSJR efficiently mines and fuses information from multiple modalities to boost classification capabilities.

In the experiments on the Trento and MUUFL datasets, the results of the different comparison models also demonstrated the same conclusion, that is, UACL has achieved the highest OA, AA, and Kappa. In Table III, for the Trento dataset, which has the smallest number of categories and low scene complexity, only UACL achieves an OA of more than 99%. In Fig. 9, UACL has a more consistent classification effect on the "Apple trees" category, reflecting the stronger discriminative ability of our method through the introduction of unsupervised information. In Table IV, for MUUFL datasets with significant long-tailed distribution phenomena, UACL achieves optimal accuracy for most categories (seven of 11 categories). More importantly, meaningful improvements have been achieved on OA, AA, and Kappa, which are, respectively, 5.60%, 6.72%, and 7.08% higher than the suboptimal method (CoupledCNN). This is partly because, as shown in Fig. 3, the MUUFL dataset has a distinct long-tailed class distribution, which maximizes the positive effects of the BS strategy to significantly improve performance. Fig. 10 shows that the UACL classification result is closer to the ground-truth map, such as the mixed ground surface in the lower right corner. This is attributed to the more generalized features obtained by the PUCL strategy.

Table V displays the classification performance of different models in the fourth scene (i.e., the Augsburg dataset). Due to its low spatial resolution, the classification effect is limited. But our UACL method still achieves the highest OA, AA, and Kappa, which are improved by 0.58%, 0.52%, and 0.87% compared to the suboptimal results, respectively. From the classification visualization in Fig. 11, the red residential areas have fewer isolated class points and are more consistent with the actual situation. In conclusion, both subjective and objective analyses validate the sophistication of UACL.

2) Performance With Different Training Sample Volumes: We have investigated the performance of the overall classification accuracy under various training sample quantities to validate the proposed UACL framework's robustness and effectiveness. Specifically, for each class, a fixed number of randomly selected samples, i.e., 20, 40, 60, 80, and 100, are used to train the model, and the remainder are used for testing. For the sake of fairness, all the results are averaged using the results of ten identical sets of training and test samples. As shown in Fig. 12, for all four datasets, there is a general tendency in the classification performance of all approaches, i.e., the OA is higher as the training sample volume increases. This reflects that the DL-based models' performance is strongly dependent on the number of labeled samples, which is one of the starting points for exploring unsupervised information utilization. More importantly, UACL always performs the highest OA for each training sample volume case of the four datasets. In particular, the advantage of our UACL is more pronounced if the training sample size is scarce (e.g., 20 per category). This is attributed to the design of the BS and the exploitation of the unlabeled sample information.

IV. Conclusion

Dedicated to alleviating the labeled sample dependence of the DL-based joint classification methods for multimodal RS data, a novel UACL model is proposed in this article. Specifically, we design an MSJR pipeline and a PUCL strategy. First, in MSJR, the multimodal spectral–spatial information is effectively fused, while multilevel class probability estimation and deeper feature mapping are achieved. In PUCL, to properly alleviate the negative effect of long-tailed distribution, we conduct a BS in the unlabeled sample set and feed them to the training set. Subsequently, unlabeled samples are divided into reliable and unreliable parts based on the predicted probability distribution. Reliable pseudo-labeled samples are used to push the network to learn discriminative features, which can minimize intra-class distance and maximize the inter-class distance. For hard samples corresponding to unreliable pseudo-labels, further positive and negative sample relationship is measured adaptively with soft mask constraints. Within the overall loss condition, the supervised classification and the unsupervised HCL process facilitate each other. We have carried out relatively comprehensive experiments and discussions on four

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commonly used multimodal RS datasets to prove the effectiveness and advancement of the designed UACL. Through objective and subjective analyses, the proposed approach is proved to be superior to other state-of-the-art models.

Furthermore, the UACL has shortcomings that require further investigation. For example, the selection of the number of unlabeled samples, taking into account both computational overhead and accuracy, needs to be studied more carefully. Besides, the mining of weakly supervised information, such as inexact label information, is one of the future research directions.

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